# Failure Case F-005: Diversity Hard-Blocking

Severity: HIGH | Version: v0.1-prototype | Date: 2026-04-14

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| **Trigger** | All 5 levels were solvable, but one pair had Jaccard similarity > 0.5 (diversity violation). |
| **What happened** | QA flagged the diversity issue as a hard failure. The pipeline burned all 3 debug cycles trying to fix a cosmetic diversity problem, even though every level was already solvable.<br><br>The Debugger cannot fix diversity -- it only adjusts tile positions within a single level. The Developer correctly classified this as a design flaw (D5) and routed to Level Designer, but even after redesign the similarity threshold kept triggering.<br><br>Result: 3 wasted debug cycles on a non-critical issue while all levels were playable. |
| **Root cause** | The routing system correctly classified diversity violations as design flaws (D5), but the pipeline's iteration logic did not distinguish between solvability failures (critical) and diversity failures (soft). The route_after_qa() function only checked failed_level_ids without considering failure severity. |
| **Fix applied** | Added soft-fail logic in route_after_qa(): if all levels are solvable and at least 1 debug cycle has been attempted, accept the result and finalize. Diversity is now treated as a soft constraint, not a hard blocker. |
| **Governance lesson** | Not all failures are equal. A system that treats cosmetic issues with the same urgency as critical bugs will waste its iteration budget. Severity-aware routing is necessary for efficient use of bounded resources. |